

WHIDBEY ISLAND NAS, WA, USA

WMO: 720749

Lat: 48.350N

Lon: 122.667W

Elev: 14

StdP: 101.15

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 24255

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 12	(b) -4.1	(c) -2.4	(d) -11.3	(e) 1.4	(f) -1.5	(g) -8.4	(h) 1.8	(i) 0.0	(j) 16.1	(k) 8.4	(l) 14.9	(m) 8.2	(n) 2.4	(o) 100	(p) 0.598

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 7.4	(c) 22.3	(d) 16.1	(e) 20.9	(f) 15.6	(g) 19.2	(h) 14.8	(i) 16.8	(j) 20.8	(k) 16.0	(l) 19.8	(m) 15.4	(n) 18.7	(o) 3.6	(p) 260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.0	10.7	17.9	14.1	10.0	17.3	13.6	9.7	17.0	46.6	20.9	44.7	19.8	42.9	18.8	20.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 13.6	(b) 11.8	(c) 10.2	(e) -7.4	(f) 26.4	(g) 2.5	(h) 1.9	(i) -9.1	(j) 27.7	(k) -10.6	(l) 28.8	(m) -12.0	(n) 29.9	(o) -13.8	(p) 31.3		
(4) 13.6	(5) 11.8	(6) 10.2	(7) -7.4	(8) 26.4	(9) 2.5	(10) 1.9	(11) -9.1	(12) 27.7	(13) -10.6	(14) 28.8	(15) -12.0	(16) 29.9	(17) -13.8	(18) 31.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.2	5.7	5.7	7.6	9.4	11.6	13.8	15.2	15.4	13.8	10.5	7.6	5.2		
	DBStd	4.46	3.45	2.91	2.57	2.08	2.00	1.79	1.84	1.78	2.10	2.53	3.39	3.34		
	HDD10.0	641	137	121	83	34	7	0	0	0	1	23	84	151		
	HDD18.3	2992	393	353	333	268	208	138	98	92	137	242	322	407		
	CDD10.0	697	3	2	8	17	57	113	162	168	114	40	12	2		
	CDD18.3	6	0	0	0	0	0	1	2	2	1	0	0	0		
	CDH23.3	21	0	0	1	0	1	2	9	6	2	0	0	0		
	CDH26.7	1	0	0	0	0	1	0	1	0	0	0	0	0		
(14) Wind	WSAvg	3.7	4.4	4.1	4.6	4.0	3.3	3.2	2.9	2.6	2.5	3.6	4.7	4.8		
(15) Precipitation	PrecAvg	495	62	41	40	37	33	29	19	22	30	43	65	65		
	PrecMax	778	155	83	92	65	69	69	54	53	107	147	225	165		
	PrecMin	345	9	14	8	11	10	0	0	0	1	7	17	11		
	PrecStd	96	32	16	17	14	15	17	14	18	25	28	40	30		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.7	14.1	17.1	17.8	19.7	22.9	24.4	24.0	22.3	19.0	17.3	13.8		
		MCWB	10.8	10.0	11.3	12.2	14.2	16.1	17.2	16.6	15.9	14.0	13.0	10.5		
	2%	DB	12.4	12.4	14.1	15.8	17.6	20.3	22.2	22.1	20.2	17.3	14.6	11.8		
		MCWB	9.9	9.2	10.1	11.0	13.3	15.0	16.2	16.2	15.1	13.4	12.0	9.6		
	5%	DB	11.3	11.1	12.7	14.1	16.4	18.2	20.4	20.3	18.8	15.8	13.1	10.7		
		MCWB	9.0	8.5	9.4	10.2	12.6	14.0	15.5	15.5	14.6	12.6	10.9	8.5		
	10%	DB	10.2	9.9	11.6	13.0	15.2	17.2	18.8	18.9	17.6	14.4	12.2	9.7		
		MCWB	8.1	7.7	8.7	9.7	11.9	13.5	14.8	15.0	14.1	11.8	10.1	7.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.1	11.3	12.3	13.0	15.2	16.9	17.9	17.5	17.0	15.6	13.9	11.5		
		MCDB	13.2	13.0	15.2	16.5	17.9	21.1	23.2	21.5	20.2	18.0	16.2	13.0		
	2%	WB	10.5	10.0	10.8	11.8	13.9	15.6	16.7	16.7	15.8	14.1	12.4	10.0		
		MCDB	11.8	11.6	13.0	14.5	16.9	19.4	21.3	20.6	18.9	16.1	14.2	11.4		
	5%	WB	9.3	8.9	9.9	10.9	13.0	14.5	15.8	15.9	15.1	13.0	11.4	8.8		
		MCDB	10.8	10.5	12.2	13.4	15.8	17.8	19.7	19.6	18.0	15.1	13.1	10.3		
	10%	WB	8.3	7.9	9.0	10.2	12.3	13.8	15.0	15.3	14.5	12.2	10.2	7.9		
		MCDB	9.7	9.4	11.2	12.7	14.8	16.7	18.4	18.6	17.2	14.0	11.8	9.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.4	6.5	6.8	6.9	6.6	6.8	7.2	7.4	7.6	6.9	5.9	5.2		
		MCDBR	5.9	7.4	8.1	8.7	8.3	8.9	9.7	9.4	9.0	7.8	5.8	5.0		
	5% WB	MCWBR	4.2	5.0	5.2	5.2	5.1	5.1	5.1	4.9	5.1	5.0	4.1	3.9		
		MCWBR	5.1	6.0	7.2	8.0	7.8	8.3	9.0	8.5	7.8	6.4	5.2	4.5		
(40) Clear-Sky Solar Irradiance	taub	taub	0.317	0.314	0.327	0.349	0.372	0.355	0.346	0.351	0.338	0.328	0.320	0.313		
		taud	2.536	2.560	2.540	2.469	2.417	2.486	2.539	2.527	2.558	2.576	2.537	2.502		
	Ebn at Noon	Ebn at Noon	764	856	897	904	890	907	910	891	873	826	750	715		
		Edh at Noon	61	72	86	102	111	105	98	95	84	70	60	56		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	0.94	1.82	2.83	4.27	5.45	5.96	6.53	5.51	3.94	2.17	1.12	0.75		
	RadStd	RadStd	0.12	0.24	0.33	0.40	0.43	0.57	0.46	0.36	0.33	0.24	0.17	0.10		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+2
(47) Regional (70 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page